Total No. of Questions : 12]

SEAT No. :

PA-170

[Total No. of Pages : 3

[5927] - 53

B.E. (Civil)

DAMS AND HYDRAULIC STRUCTURES

(2015 Pattern) (Semester - II) (401007)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12,***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of electronic non-programmable calculator is allowed.***
5) ***Assume suitable data if necessary.***

Q1) a) Write a short note on Dams and Social Issues. **[3]**

b) Explain types of measurements in Dams safety. **[3]**

OR

Q2) a) Differentiate between Large Dam and Small Dam. **[3]**

b) Write Short note on Piezometers and its working principle. **[3]**

Q3) a) Explain various forces acting on Gravity Dam. **[5]**

b) State advantages and limitations of butters dam. **[3]**

OR

Q4) a) What are the Modes of failure of gravity dam? What is Middle third rule? **[4 + 2]**

b) What are the factors affecting selection of arch dam? **[2]**

P.T.O.

Q5) Correlation between jump height and tail water depth. **[6]**

OR

Q6) What is a spillway gate? Briefly explain any two types of gates. **[2 + 2 + 2]**

Q7) a) Differentiate between a weir and a barrage. **[6]**

b) Explain Classification of an earth dam based on materials, method of construction. **[6]**

c) Write the procedure of stability analysis of an earth dam by Swedish slip circle method. **[6]**

OR

Q8) a) Briefly outline Khosla's theory on the design of weir on permeable foundations. **[6]**

b) How would you proceed to determine the phreatic line through homogenous earthen dams provided **[3 + 3]**

i) with a horizontal filter.

ii) without a horizontal filter.

c) Draw a neat layout of diversion head works and indicate the various components along with explain their functions in brief. **[6]**

Q9) a) What is a canal? Explain three types of canals based on function. **[8]**

b) Design an irrigation channel in alluvial soil from following data using Lacey's theory : **[8]**

Discharge 20.0 cumec ;

Lacey's silt factor 1.0 ;

Side slope = ½ : 1

OR

Q10) a) What is a Canal Fall? Discuss the necessity of it and also explain any two types of falls. **[4 + 2 + 2]**

b) Design an irrigation channel by Kennedy's theory to carry a discharge of 5 cumecs. Take m = 1.0, N = 0.0225 and B/D = 4.4. **[8]**

Q11) a) Enlist objective of river training works. State and explain any three river training works in detail. **[2 + 6]**

b) What are the cross drainage works? Explain aqueduct with neat sketch. **[2 + 6]**

OR

Q12) a) Classification of Cross Drainage works according to drain over canal and canal over drain. **[4 + 4]**

b) Write a short note on : **[2 + 3 + 3]**

i) Guide Bunds.

ii) Attracting groyne.

iii) Deflecting groyne.